EDS 222: Homework 1

Foundations of Inference

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2026-10-22

```
library(tidyverse)
set.seed(222)
theme_set(theme_bw())
```

1. Conduct a permutation test

Conduct a permutation test to investigate the following question:

> Do homes built pre-1970 have higher average lead soil concentrations?

To perform this test, you'll need the `exide-property-covariates.csv` file found in `/courses/EDS222/data`. This file contains the year each property was built.

> **Note**
>
> The question is about *average lead soil concentrations*. You'll need to first get the average lead soil concentration for each property, then join your result to the property covariate data to get the year the house was built.

Permutation Test: Average Lead Soil Concentrations

Comparing Homes Built Before 1970 and Homes Built in 1970 or Later

Question

Do homes built pre-1970 have higher average lead soil concentrations?

Hypothesis

Ho: There is no difference in lead concentration between homes built before 1970 and homes built after 1970. Ha: Homes built before 1970 have higher lead concentrations compared to homes built after 1970.

```
soil_cores<-read_csv("exide-soil-cores.csv")

glimpse(soil_cores)
```

```
Rows: 3,839
Columns: 7
$ property_id   <chr> "ENA-01", "ENA-01", "ENA-01", "ENA-01", "ENA-01", "ENA-0…
$ sample_id     <chr> "ENA-01-10D-PA-03", "ENA-01-1D-03", "ENA-01-2D-03", "ENA…
$ sample_date   <date> 2014-09-05, 2014-09-05, 2014-09-05, 2014-09-05, 2014-09…
$ location_code <chr> "YD", "YD", "YD", "YD", "YD", "YD", "YD", "YD", "YD", "Y…
$ pb_ppm        <dbl> 78.1, 150.0, 132.0, 130.0, 108.0, 114.0, 134.0, 146.0, 1…
$ longitude     <dbl> -118.1910, -118.1909, -118.1909, -118.1909, -118.1910, -…
$ latitude      <dbl> 34.02617, 34.02619, 34.02617, 34.02618, 34.02620, 34.026…
```

```
prop_cv <- read_csv("exide-property-covariates.csv")

glimpse(prop_cv)
```

```
Rows: 324
Columns: 3
$ property_id       <chr> "ENA-01", "ENA-02", "ENA-03", "ENA-04", "ENA-05", "E…
$ year_built        <dbl> 1920, 1921, 1912, 1921, 1905, 1948, 1922, 1939, 1924…
$ dist_to_freeway_m <dbl> 539, 202, 242, 326, 93, 264, 205, 311, 184, 181, 475…
```

```
sc_yr <- prop_cv |>
  filter(!is.na(year_built))
```

```
pi_means<- soil_cores |>
    summarize(mean_pb_ppm = mean(pb_ppm), .by = property_id)
```

```
# Joining datasets
# Property id is shared key

sc_yr <- pi_means|>
```

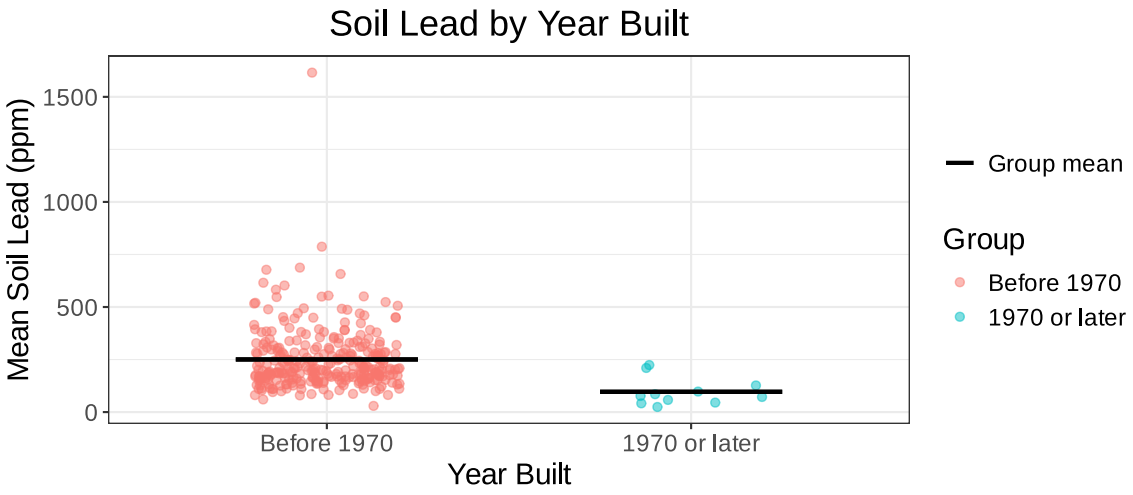
```r
  inner_join(sc_yr, join_by(property_id)) |>
  select(property_id, mean_pb_ppm, year_built)
```

```r
# Creating groups for homes built before 1970 and in 1970 or later
sc_yr <- sc_yr |>
  mutate(
    year_group = if_else(year_built < 1970, "pre", "post")
  )

# Calculating observed difference in means
obs_summ <- sc_yr |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE), .by = year_group)

obs_yr_dm <- obs_summ$mean_pb_ppm[obs_summ$year_group == "pre"] -
  obs_summ$mean_pb_ppm[obs_summ$year_group == "post"]
```

```r
# Visualizing raw data by year built
ggplot(sc_yr, aes(x = year_group, y = mean_pb_ppm, color = year_group)) +
  geom_point(size = 1.75, alpha = 0.5,
             position = position_jitter(width = 0.2, height = 0)) +
  stat_summary(
    aes(linetype = "Group mean"),
    fun = mean, fun.min = mean, fun.max = mean,
    geom = "errorbar", width = 0.5,
    color = "black", linewidth = 1,
    key_glyph = "path"
  ) +
  scale_x_discrete(limits = c("pre", "post"),
                   labels = c("pre" = "Before 1970", "post" = "1970 or later")) +
  scale_color_discrete(limits = c("pre", "post"),
                           labels = c("pre" = "Before 1970", "post" = "1970 or
later")) +
  theme(text = element_text(size = 15),
        plot.title = element_text(hjust = 0.5)) +
  labs(
    x = "Year Built",
    y = "Mean Soil Lead (ppm)",
    title = "Soil Lead by Year Built",
    color = "Group",
    linetype = NULL
  )
```




```
# Calculating shuffled group means by year
shuff_summ <- sc_yr |>
  mutate(year_group = sample(year_group)) |>
  summarize(
    mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE),
    .by = year_group
  )

# Shuffling/Randomizing predictor variable
# Breaks connection between year and lead concentration
# Recalculates statistic

shuff_diff_means <-
  shuff_summ$mean_pb_ppm[shuff_summ$year_group == "pre"] -
  shuff_summ$mean_pb_ppm[shuff_summ$year_group == "post"]

shuff_diff_means
```

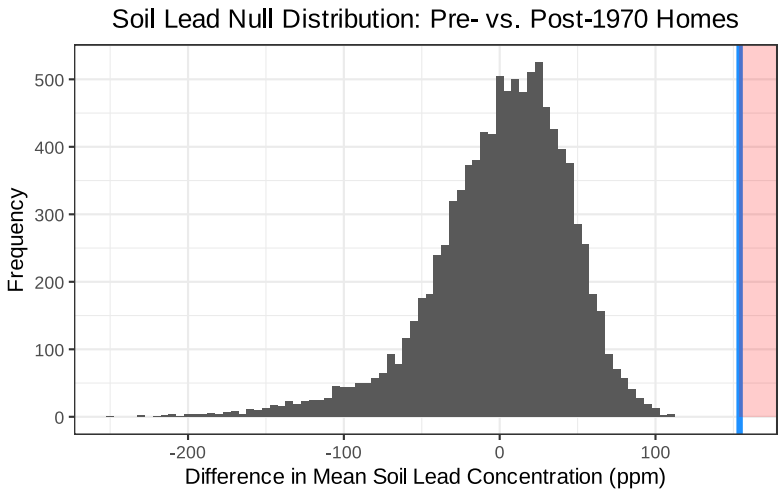
```
[1] -13.56897
```

```
# Creates NULL distribution
y_null_dist <- replicate(
  10000,
  {
    shuff_summ <- sc_yr |>
      mutate(year_group = sample(year_group)) |>
     summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE), .by = year_group)
```

```
    shuff_summ$mean_pb_ppm[shuff_summ$year_group == "pre"] -
      shuff_summ$mean_pb_ppm[shuff_summ$year_group == "post"]
  }
)
```

Results

```
# Graphs Permutation NUll Distribution
tibble(diff_means = y_null_dist) |>
  ggplot(aes(diff_means)) +
  geom_histogram(binwidth = 5) +
  labs(
    x = "Difference in Mean Soil Lead Concentration (ppm)",
    y = "Frequency",
    title = "Soil Lead Null Distribution: Pre- vs. Post-1970 Homes"
  )+ theme(
  plot.title = element_text(hjust = 0.5))+
   geom_vline(xintercept = obs_yr_dm, color = "dodgerblue", linewidth = 1.5) +
annotate("rect", xmin = obs_yr_dm, xmax = Inf, ymin = 0, ymax = Inf, fill =
"red", alpha = 0.2)
```




```
pval_yr <- mean(y_null_dist > obs_yr_dm)

pval_yr
```

```
[1] 0
```

```
obs_yr_dm
```

```
[1] 153.9808
```

The observed difference in mean soil lead between homes built before 1970 and homes built after 1970 is 153.98 ppm. In 10,000 permutations, none produced a difference as large as 153.98 ppm, so the estimated p-value is approximately 0. The shaded region marks the p-value area. This observed difference is much larger than the differences we would expect from random chance alone.

Conclusion

Since the p-value is less than our threshold of 0.05, we reject the null hypothesis that there is no difference in lead concentration between homes built before 1970 and homes built after 1970. There is enough evidence to support the alternative hypothesis that homes built before 1970 have higher lead concentrations compared to homes built after 1970. The large difference in group sizes, 303 homes before 1970 and only 11 after, could impact our inferences.

2. Estimate a confidence interval

Estimate a confidence interval to investigate the following question:

> How much is the difference in average lead soil concentrations between properties within 1km of a freeway and those farther away?

```
sc_frw <- pi_means|>
  inner_join(prop_cv, join_by(property_id)) |>
  select(property_id, mean_pb_ppm, dist_to_freeway_m) |>
  mutate(km_frw = dist_to_freeway_m/1000)
```

```
# Creating groups for distance from freeway, less than or equal to 1 km and more
than 1 km
sc_frw <- sc_frw |>
  mutate(
    frw_group = if_else(km_frw <= 1, "less", "more")
  )
```

```
# Filtering groups
less_1km <- sc_frw |>
  filter(km_frw <= 1)
more_1km <- sc_frw |>
  filter(km_frw > 1)
```

```r
#  Calculating observed group means
obs_summ <- sc_frw |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE), .by = frw_group)

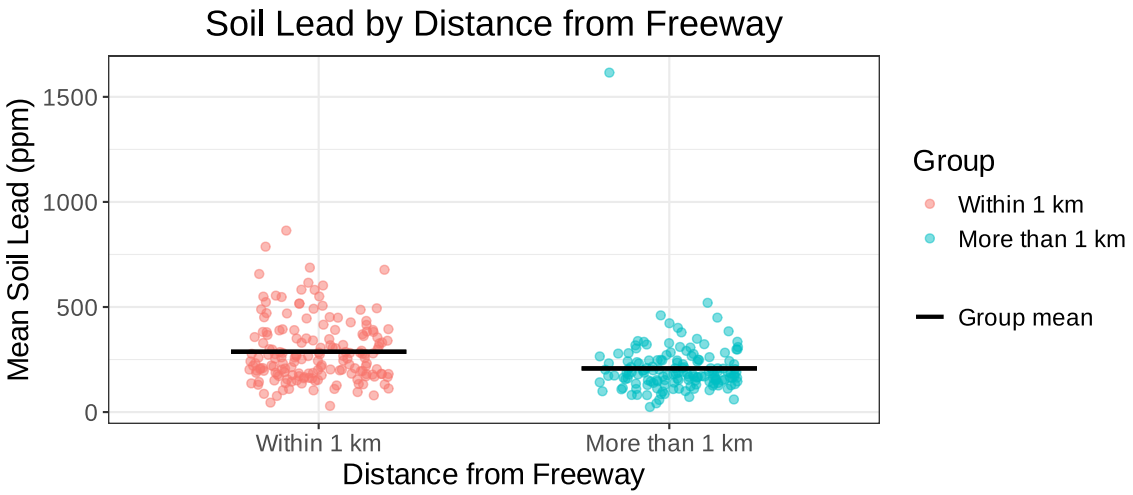
ob_km_dm <- obs_summ$mean_pb_ppm[obs_summ$frw_group == "less"] -
  obs_summ$mean_pb_ppm[obs_summ$frw_group == "more"]

ob_km_dm
```

```
[1] 79.32728
```

```r
# Visualizing raw data by freeway distance group

# Visualizing raw data by freeway distance group
ggplot(sc_frw, aes(x = frw_group, y = mean_pb_ppm, color = frw_group)) +
  geom_point(size = 1.75, alpha = 0.5,
             position = position_jitter(width = 0.2, height = 0)) +
  stat_summary(
    aes(linetype = "Group mean"),
    fun = mean, fun.min = mean, fun.max = mean,
    geom = "errorbar", width = 0.5,
    color = "black", linewidth = 1,
    key_glyph = "path"
  ) +
   scale_x_discrete(labels = c("less" = "Within 1 km", "more" = "More than 1
km")) +
  scale_color_discrete(labels = c("less" = "Within 1 km", "more" = "More than 1
km")) +
  theme(text = element_text(size = 15),
        plot.title = element_text(hjust = 0.5)) +
  labs(
    x = "Distance from Freeway",
    y = "Mean Soil Lead (ppm)",
    title = "Soil Lead by Distance from Freeway",
    color = "Group",
    linetype = NULL
  )
```




```
# 1 bootstrap sample
# Sample for properties within 1 km
less_km <- less_1km |>
  slice_sample(
    n = nrow(less_1km),
    replace = TRUE
  )
# Sample for properties farther than 1 km
more_km <- more_1km |>
  slice_sample(
    n = nrow(more_1km),
    replace = TRUE
  )
```

```
# Calculating difference
less_sp <- less_km |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE))
more_sp <- more_km |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE))
frw_mean_diff <-
  less_sp$mean_pb_ppm - more_sp$mean_pb_ppm
frw_mean_diff
```

```
[1] 98.20663
```

```
frw_dist <- replicate(
  10000,
```

```
  {
    less_sample <- less_1km |>
  slice_sample(
    n = nrow(less_1km),
    replace = TRUE
  )
  more_sample <- more_1km |>
  slice_sample(
    n = nrow(more_1km),
    replace = TRUE
  )
less_sp <- less_sample |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE))
more_sp <- more_sample |>
  summarize(mean_pb_ppm = mean(mean_pb_ppm, na.rm = TRUE))

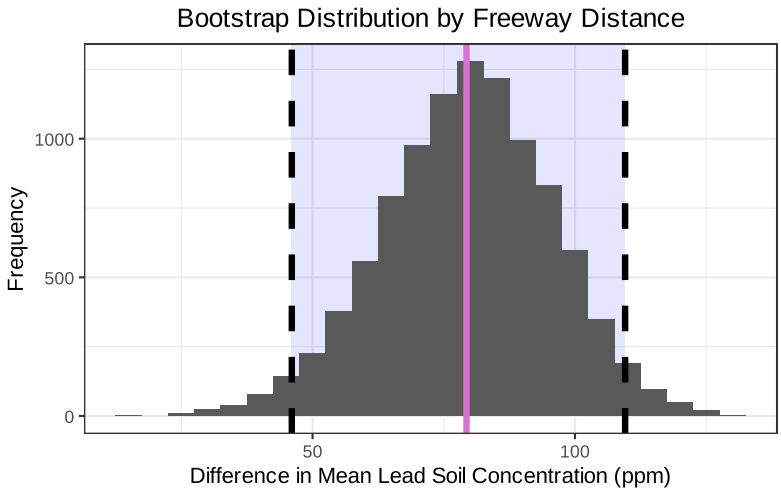
less_sp$mean_pb_ppm - more_sp$mean_pb_ppm
  }
)
```

Results

```
ci <- quantile(frw_dist, probs = c(0.025, 0.975))
print(ci)
```

```
    2.5%    97.5%
46.04037 109.55488
```

```
tibble(diff_means = frw_dist) |>
  ggplot(aes(diff_means)) +
  annotate("rect", xmin = ci[1], xmax = ci[2], ymin = 0, ymax = Inf,
             fill = "blue", alpha = 0.1) +
  geom_histogram(binwidth = 5) +
  labs(
    x = "Difference in Mean Lead Soil Concentration (ppm)",
    y = "Frequency",
    title = "Bootstrap Distribution by Freeway Distance"
  )+ theme(
  plot.title = element_text(hjust = 0.5))+
  geom_vline(xintercept = ob_km_dm, color = "orchid", linewidth = 1.5) +
  geom_vline(xintercept = ci, linetype = "dashed", linewidth = 1.5)
```



The observed difference in mean soil lead concentration is 79.3 ppm. The bootstrap distribution is centered around this value.

The 95% confidence interval for the difference in mean lead soil concentration for homes within 1 km of a freeway and those farther away is 46.04 ppm to 109.55 ppm.

Conclusion

The plausible values for the difference in population means are between 46.04 ppm and 109.55 ppm.

This provides evidence that properties within 1 km of a freeway had average lead soil concentrations that are about 79 ppm higher than properties that are farther than 1 km from the freeway.

If we repeatedly sampled and built an interval, about 95% of those intervals would contain the true difference in mean soil lead between properties within 1 km of a freeway and those farther away.

3. Simulate a negative result

Recall that a p-value is the probability of observing your sample *if* the null hypothesis is true. Therefore, if you draw a sample from two groups that part of the same population, you should get a p values less than 0.05 only 5% of the time. In other words, a negative result (no actual difference) will yield a significant p-value at the 0.05 threshold in 5% of samples. This is known as the **false positive rate**.

To simulate a negative result, perform the following.

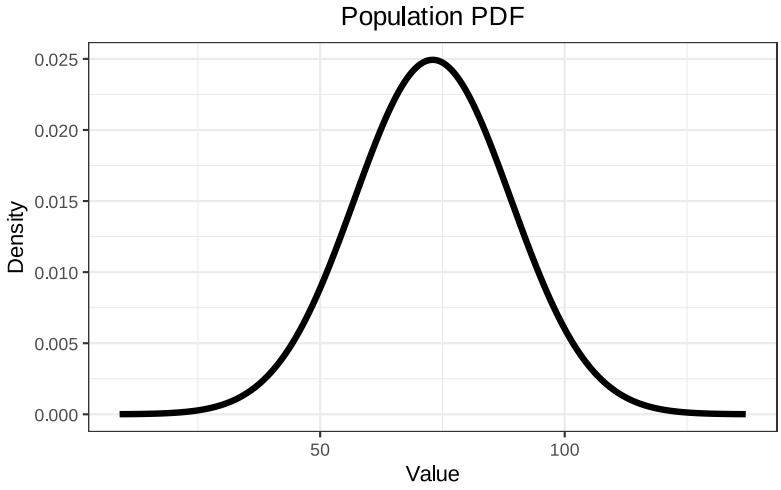
1. Define a normally-distributed population. Pick any mean and standard deviation.

Population Mean = 73 Population SD = 16

```
normal_sample<- tibble(x = rnorm(50, mean = 73, sd = 16))
normal_sample
```

```
# A tibble: 50 × 1
       x
   <dbl>
 1  79.1
 2  76.8
 3  46.1
 4  85.3
 5  53.9
 6  68.7
 7  56.8
 8  96.3
 9  68.2
10  60.1
# ℹ 40 more rows
```

```
tibble(
  x = seq(9, 137, length.out = 1000),
  pdf = dnorm(x, mean = 73, sd = 16)
) |>
  ggplot(aes(x, pdf)) +
  geom_line(linewidth = 1.5) +
  theme(
  plot.title = element_text(hjust = 0.5))+
  labs(x = "Value", y = "Density", title = "Population PDF")
```



2. Draw a sample of 50 observations from this population. Assign the first 25 to group A and the second 25 to group B.

```
normal_sample <- normal_sample |>
  mutate(
      group = if_else(1:50 <=25, "A", "B"))
```

3. Conduct a permutation test to investigate whether the mean of group A is greater than the mean of group B.

```
means<- normal_sample |>
    summarize(mean= mean(x), .by = group)
  means
```

```
# A tibble: 2 × 2
  group  mean
  <chr> <dbl>
1 A      75.5
2 B      71.6
```

```
# Filtering groups
a <- normal_sample |>
  filter(group == "A")
b <- normal_sample |>
  filter(group == "B")

#  Calculating observed group means
a_mean <- a |>
  summarize(mn_a = mean(x, na.rm = TRUE))
a_mean
```

```
# A tibble: 1 × 1
   mn_a
  <dbl>
1  75.5
```

```
b_mean <-b |>
  summarize(mn_b = mean(x, na.rm = TRUE))
b_mean
```

```
# A tibble: 1 × 1
   mn_b
```

```
  <dbl>
1  71.6
```

```
ob_ab_dm <- a_mean$mn_a - b_mean$mn_b
```

```
# Calculating group means by group
shuff_sm <- normal_sample |>
  mutate(group = sample(group)) |>
  summarize(
    x = mean(x, na.rm = TRUE),
    .by = group
  )
```

```
# Shuffling/Randomizing predictor variable
# Recalculates statistic
shuff_mn_diff <-
  shuff_sm$x[shuff_sm$group == "A"] -
  shuff_sm$x[shuff_sm$group == "B"]
shuff_mn_diff
```

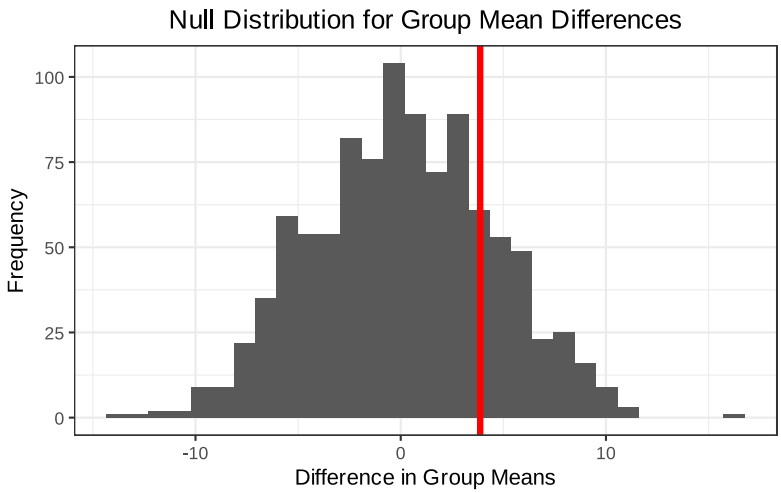
```
[1] 1.837235
```

```
# Creates permuttion NULL distribution

ab_null_dist <- replicate(
  1000,
  {
    shuff_sm <- normal_sample |>
      mutate(group = sample(group)) |>
      summarize(
        x = mean(x, na.rm = TRUE),
        .by = group
        )
        shuff_sm$x[shuff_sm$group == "A"] -
          shuff_sm$x[shuff_sm$group == "B"]
          }
          )
```

```
#Graphs permutation NUll Distribution
```

```r
tibble(diff_means = ab_null_dist) |>
  ggplot(aes(diff_means)) +
  geom_histogram() +
  labs(
    x = "Difference in Group Means",
    y = "Frequency",
    title = "Null Distribution for Group Mean Differences",
    ) +
      theme(
  plot.title = element_text(hjust = 0.5))+
      geom_vline(xintercept = ob_ab_dm, color = "red", linewidth = 1.5)
```



4. Record your p-value.

```r
pval_ab <- mean(ab_null_dist > ob_ab_dm )
print(pval_ab)
```

```
[1] 0.205
```

5. Repeat the simulation 1,000 times.

```r
  # each pass through the braces of the outer replicate() is one full experiment
n_sims <- 1000
n_perms <- 1000

p_vals <- replicate(
  n_sims,
  {
```

```
    normal_sample <- tibble(x = rnorm(50, mean = 73, sd = 16)) |>
      mutate(group = if_else(1:50 <= 25, "A", "B"))

    a_mean <- normal_sample |>
      filter(group == "A") |>
      summarize(mn_a = mean(x, na.rm = TRUE))
    b_mean <- normal_sample |>
      filter(group == "B") |>
      summarize(mn_b = mean(x, na.rm = TRUE))
    ob_ab_dm <- a_mean$mn_a - b_mean$mn_b

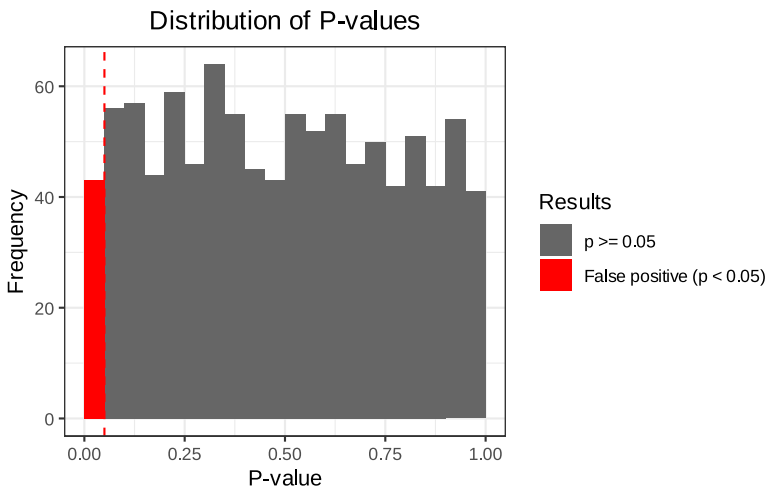
    ab_null_dist <- replicate(
      n_perms,
      {
        shuff_sm <- normal_sample |>
          mutate(group = sample(group)) |>
          summarize(x = mean(x, na.rm = TRUE), .by = group)
        shuff_sm$x[shuff_sm$group == "A"] - shuff_sm$x[shuff_sm$group == "B"]
      }
    )

    (sum(ab_null_dist >= ob_ab_dm) + 1) / (n_perms + 1)
  }
)

all_pvals <- tibble(p_value = p_vals)
```

```
# Visualizing
all_pvals |>
   ggplot(aes(x = p_value, fill = p_value < 0.05)) +
   geom_histogram(binwidth = 0.05, boundary = 0) +
   scale_fill_manual(values = c("grey40", "red"),
                     labels = c("p >= 0.05", "False positive (p < 0.05)"),
                     name = "Results") +
  geom_vline(
    xintercept = 0.05,
    linetype = "dashed",
    color = "red"
  ) + theme(
  plot.title = element_text(hjust = 0.5))+
  labs(
    x = "P-value",
```

```
    y = "Frequency",
    title = "Distribution of P-values"
  )
```



When you're done, verify that your p-value is less than 0.05 in approximately 5% (50) of your simulations.

```
mean(p_vals < 0.05)
```

```
[1] 0.043
```

4.3 % of simulations had a p-values less than 0.05, which reflects the expected false positive.